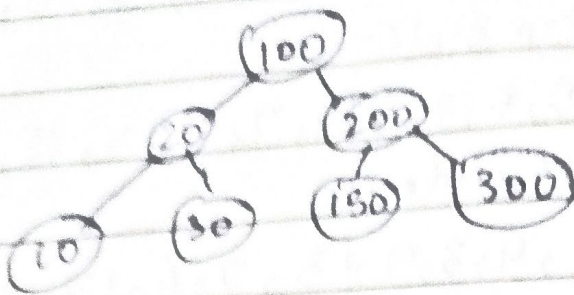


C03 - AT3 - Tree Traversal Challenges

①

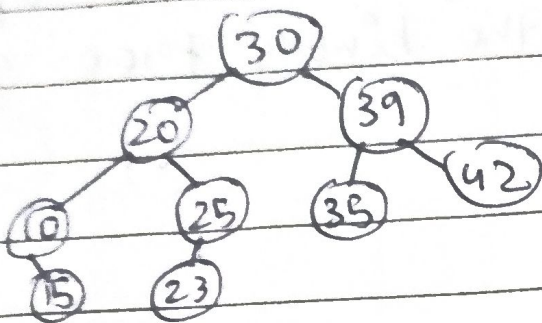


Preorder: 100, 20, 10, 30, 200, 150, 300

Inorder: 10, 20, 30, 100, 150, 200, 300

Postorder: 10, 30, 20, 150, 300, 200, 100

②

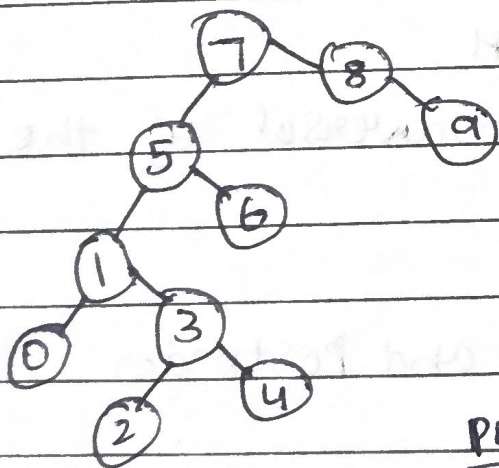


Preorder: 30, 20, 10, 15, 25, 23, 39, 35, 42

Inorder: 10, 15, 20, 23, 25, 30, 35, 39, 42

Postorder: 15, 10, 23, 25, 20, 35, 42, 39, 30

③ Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



Inorder

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

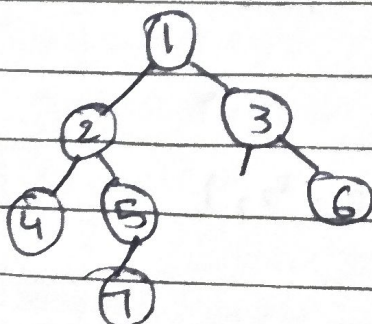
Preorder

7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Postorder

0, 2, 4, 3, 1, 6, 5, 9, 8, 7

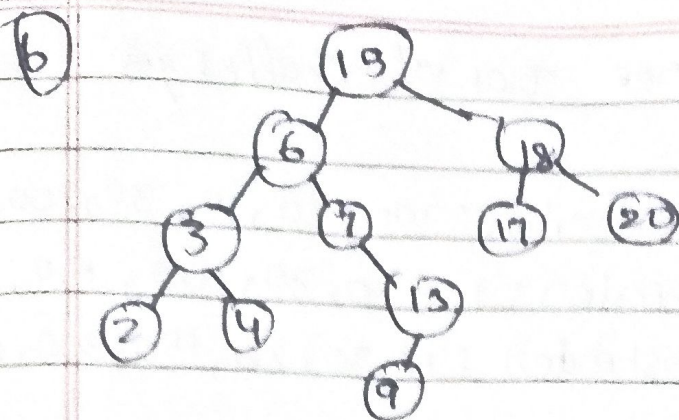
④



Preorder: 1, 2, 4, 7, 5, 3, 6

Inorder: 7, 4, 2, 5, 1, 3, 6

Postorder: 7, 4, 5, 2, 6, 3, 1



Inorder

2, 3, 4, 6, 7, 9, 13, 13, 17, 18, 20

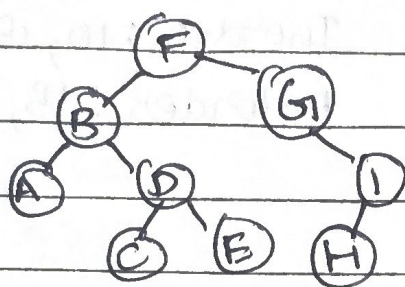
Preorder

19, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Post order

2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 19

- ⑦ What is the sequence of nodes when applying in-order traversal on the binary tree given below?



Final sequence : A, B, C, D, E, F, G, H, I

- ⑧ Consider the following Inorder and Preorder traversal of a tree

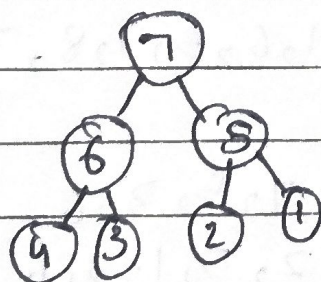
Inorder - D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder traversal of the same tree

Ans D F E B H G C A

- ⑨ Find Inorder, Preorder and Postorder



preorder

7, 6, 4, 3, 8, 2, 1

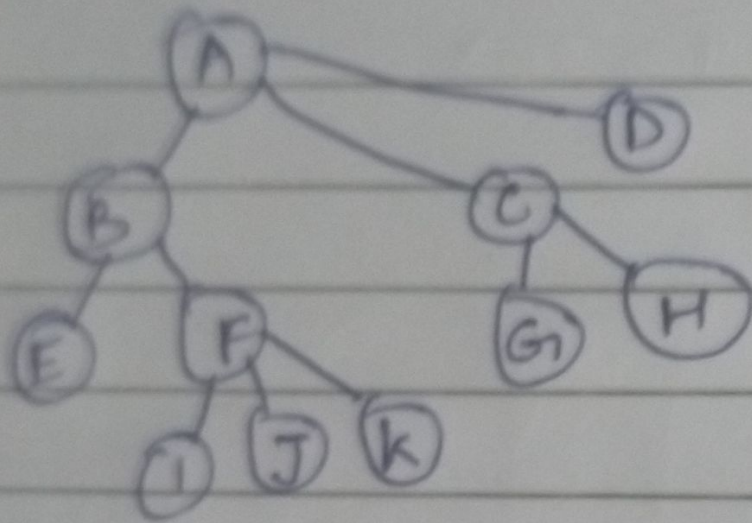
Inorder

4, 6, 3, 7, 2, 8, 1

Post order

4, 3, 6, 2, 1, 8, 7

(10)

Preorder

A, B, E, F, I, J, K, C, G, H, D

Postorder

E, I, J, K, F, B, G, H, C, D, A